

# Advanced signal processing - Multirate signal processing

## 1 Upsampling with linear interpolation

Consider a signal given by

$$x[n] = \begin{cases} -1 & n = 1 \\ 3 & n = 2 \\ 0 & \text{elsewhere} \end{cases}$$

1. Upsample the signal using  $I = 4$  and the linear interpolation kernel given by Eq. 12.48.
2. Sketch the signal before and after the upsampling and interpolation.

## 2 Downsampling is a linear operation

In example 12.1 the authors state without proof that downsampling is a linear operation.

1. Show that downsampling is indeed a linear operation.

## 3 Sample rate conversion

Explain the difference between downsampling, upsampling, decimation and interpolation.

## 4 Sample rate conversion - upsampling

Consider a signal given by

$$x[n] = \begin{cases} 2 & n = 0 \\ 1 & n = 1 \\ 0 & \text{elsewhere} \end{cases}$$

It is desired to upsample the signal with  $I = 2$  with the ideal interpolator (12.31).

1. Write out the analytical expressions and compute  $x_I[k]$  for  $k = -1, \dots, 5$ .
2. Plot the original signal and the upsampled signal and comment on the result.
3. Repeat the upsampling, but this time use linear interpolation. Compare the results of the two interpolation schemes.

## 5 Sample rate conversion

Let a signal and interpolation kernel be given by

$$x[n] = \begin{cases} 2 & n = 0 \\ -1 & n = 1 \\ 0 & \text{elsewhere} \end{cases} \quad \text{and} \quad g[n] = \begin{cases} -1 & n = -1 \\ 3 & n = 0 \\ 1 & n = 1 \\ 0 & \text{elsewhere} \end{cases}$$

1. Upsample  $x[n]$  by a factor of 3 and sketch the upsampled signal.
2. Upsample  $x[n]$  by a factor of 3 and compute the interpolated values  $x_I[1]$  and  $x_I[2]$ .
3. Discuss the performance of the  $g[n]$  interpolation kernel.

## 6 Sample rate conversion - fractional delays

In example 12.4, the authors discuss fractional delays implemented through up- and downsampling and integer delays.

1. Create a plot similar to Fig. 12.16, but where the signal has been shifted 1/3 of a sample.
2. What is the output of the system if we use an input signal containing a single impulse, e.g.,  $x[n] = \delta[n - 10]$ ?

## 7 Sample rate conversion - linear interpolation

As described in the textbook in example 12.3 linear interpolation is a very familiar interpolation technique.

1. Run the MATLAB code from example 12.3, but plot  $x[n]$  and  $x_I[n]$  on the same graph. Comment on the result.
2. Compute a mean square error of the reconstructed signal and comment on the result. Remember to discard the parts of the signal with transient distortion from the computation.

## 8 Interpolation with zero-order hold

A very simple scheme for interpolation is the zero-order hold technique, where the output is held steady, until a new sample arrives.

1. Can the zero-order hold be formulated as standard interpolation filter? If so, how, and how will it perform compared to other implementation techniques?

## 9 Interpolation with third-order hold

Problem 12.42 in the textbook is concerned with a third-order interpolator. The analytic expression for the interpolator contains the parameter  $a$  that unfortunately is not discussed at all. Further, there's a small typo in the impulse response. The correct expression is

$$g_{\text{TOH}}[n] = \begin{cases} (a+2) \left| \frac{n}{I} \right|^3 - (a+3) \left| \frac{n}{I} \right|^2 + 1, & 0 \leq |n| \leq I, \\ a \left| \frac{n}{I} \right|^3 - 5a \left| \frac{n}{I} \right|^2 + 8a \left| \frac{n}{I} \right| - 4a, & I \leq |n| \leq 2I, \\ 0, & \text{otherwise} \end{cases}$$

1. Solve problem 12.42(a) for  $a = 0.1$  and  $a = 0.5$ . Compare the magnitude response in the two cases.
2. Solve the rest of problem 12.42.

## 10 Polyphase filters

Figure 12.23 contains MATLAB code with an implementation of a polyphase FIR decimator, but the code is hardly documented and can be tricky to understand.

1. Go through the code and write a detailed explanation for each line.

## 11 Half-band filters

Consider a filter with the following system function

$$H(z) = -\frac{1}{32} \left( 1 - 9z^{-2} - 16z^{-3} - 9z^{-4} + z^{-6} \right).$$

1. Compute and plot the magnitude and phase response of the filter.
2. Show that the filter is a lowpass half-band filter.
3. Find the appropriate polyphase decomposition of the filter.
4. Check that Eq. 12.85 holds for a few selected frequencies.

## 12 Half-band filters

In chapter 12.3.1, the authors state „First, we note that the  $M/2$  zeros in the impulse response reduce the amount of computations by almost one-half“.

1. Read the blog post on:  
<https://www.wavewalkerdsp.com/2021/09/27/half-band-filter-design-exceptional-filtering-efficiency/>  
and learn how further improvements in computational efficiency is obtained.

## 13 K-band filters

1. Following the steps used in obtaining Eq. 12.84 and Eq. 12.85 for the half-band filter, show that the K-band filter satisfies Eq. 12.90 and 12.91.
2. Plot the frequency response for the K=3 case.

## 14 Perfect reconstruction filter banks

1. Prove that 12.119 follows from 12.118.

Consider two filters given by  $H_0(z) = 1 + z^{-1}$  and  $H_1(z) = 1 - z^{-1}$ .

2. Investigate if the two filters are power complementary. If not, find an appropriate scaling that enforces power complementarity.
3. Show that these filters, together with synthesis filters,  $G_0(z)$  and  $G_1(z)$ , lead to perfect reconstruction.
4. Comment on the filters ability to separate the low and high frequency bands.

## 15 Quadrature mirror filters

On page 758 in the textbook, the authors conclude: „Then the filter  $H(z)$  has the form  $H(z) = b_0 z^{-2n_0} + b_1 z^{-2n_1+1}$ , which is of limited practical usefulness.“

1. Plot the magnitude and phase response for a few selected choices of  $b_0$ ,  $b_1$ ,  $n_0$ , and  $n_1$ .
2. Do your plots support the conclusion of *limited practical usefulness*?